

SVM Mathematical Intuition

Understanding the geometry and optimization of Large Margin Classifiers.

CAMPUSX MACHINE LEARNING SERIES

Geometric Foundations

Defining the Decision Boundary in Vector Space

Defining the Separating Hyperplane

The Equation

In a d -dimensional space, a hyperplane is defined as the set of points satisfying:

$$\mathbf{w}^T \mathbf{x} + b = 0$$

Where $\mathbf{w}$ is the weight vector (perpendicular to the plane) and b is the bias constant.

Decision Rule

For a data point $\mathbf{x}_i$, our classifier predicts the label $\mathbf{y}_i$ based on the sign of the functional margin:

$$f(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x} + b)$$

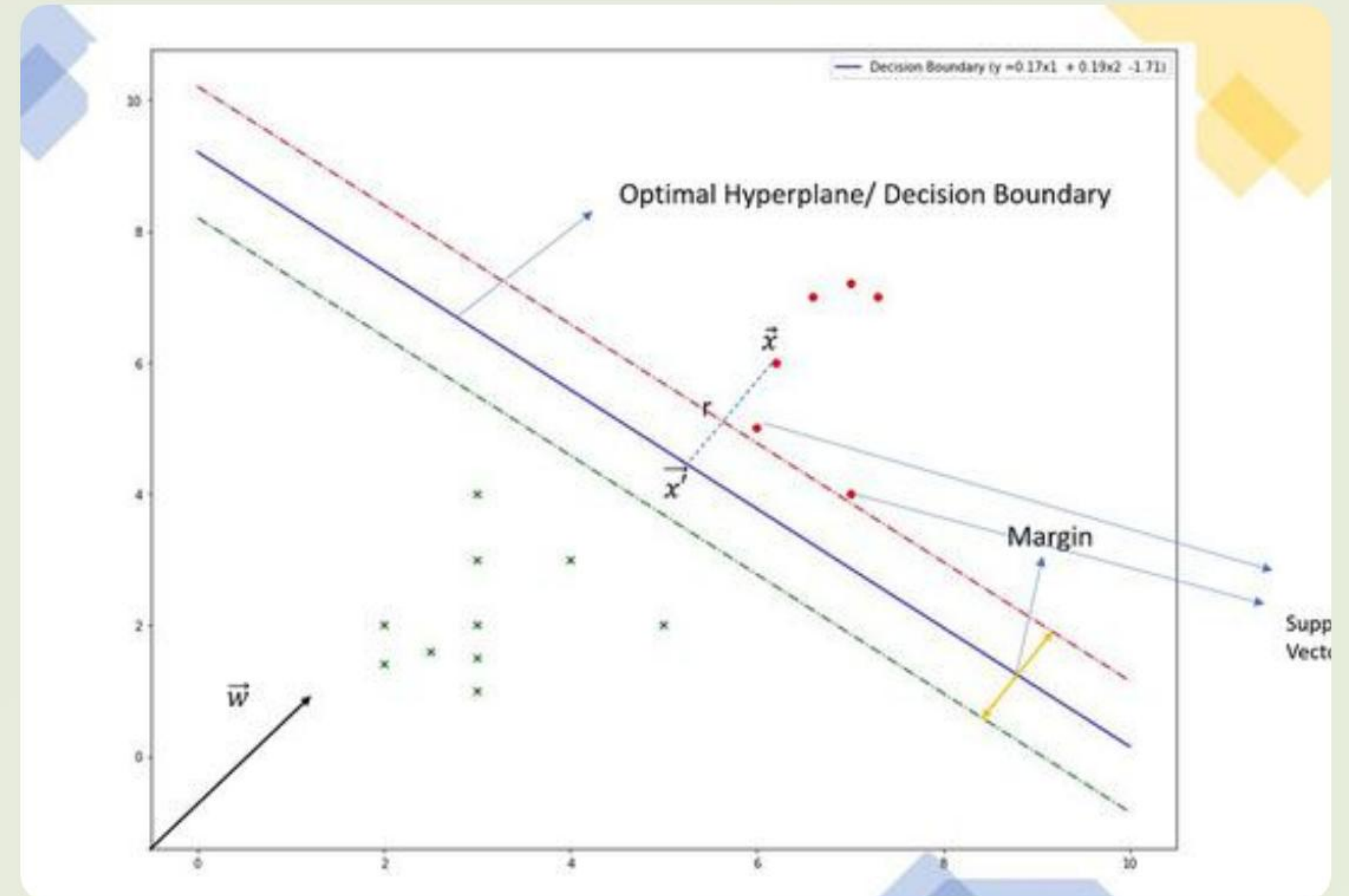
Vector Perpendicularity

The Weight Vector $\mathbf{w}$

Crucially, the vector $\mathbf{w}$ is always perpendicular to the hyperplane. This is shown by taking any two points $\mathbf{x}_1$ and $\mathbf{x}_2$ on the plane:

$$\mathbf{w}^T (\mathbf{x}_1 - \mathbf{x}_2) = 0$$

Since the difference of any two points on the plane lies in the plane, $\mathbf{w}$ must be orthogonal to every vector in the hyperplane.



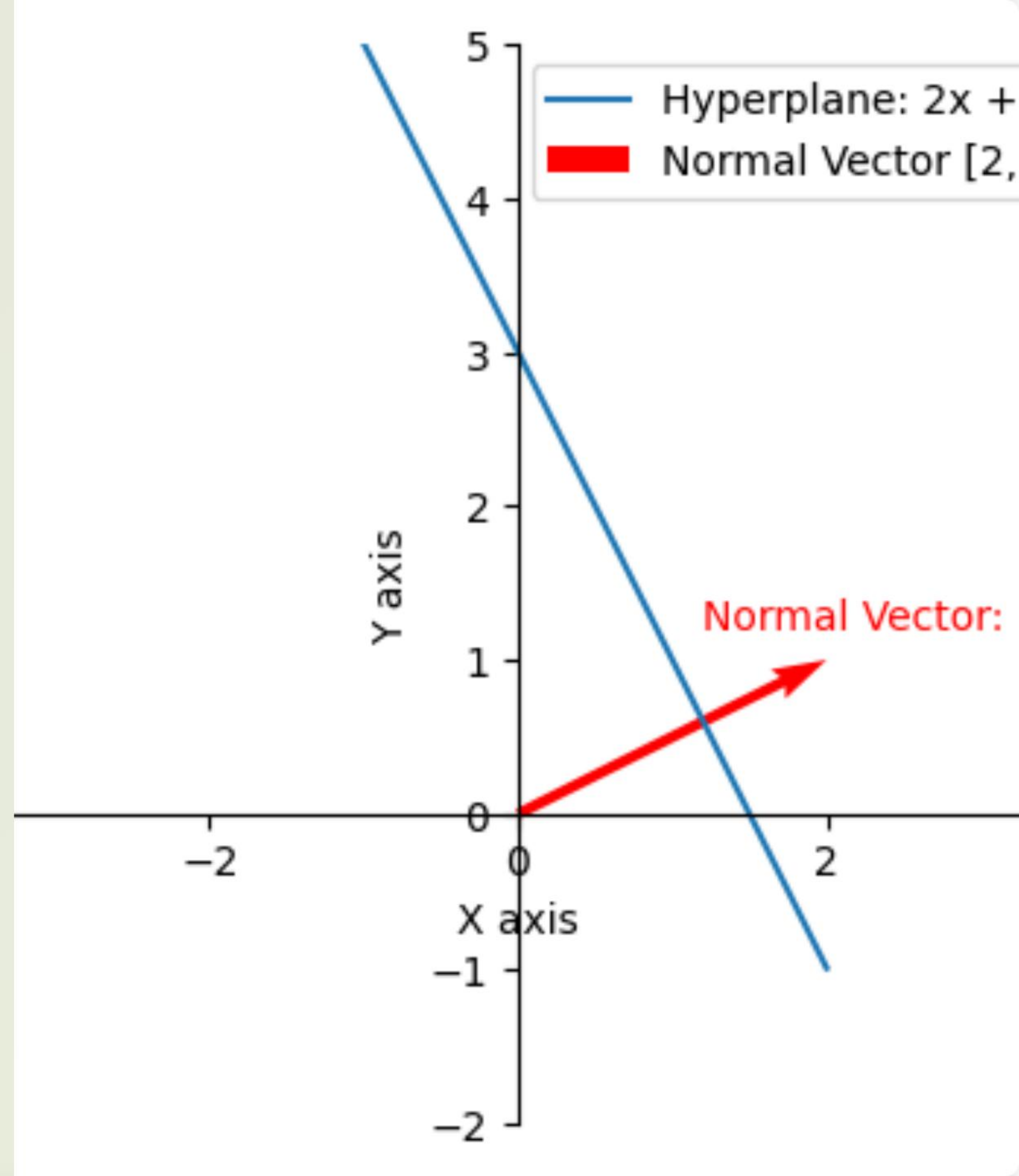
Distance from a Point

Geometric Projection

The perpendicular distance d from a point $\mathbf{x}_0$ to the hyperplane is found by projecting the vector $(\mathbf{x}_0 - \mathbf{x})$ onto the unit vector in the direction of $\mathbf{w}$.

$$\text{dist} = \frac{|\mathbf{w}^T \mathbf{x}_0 + b|}{\|\mathbf{w}\|}$$

This formula is the core of margin calculation in Support Vector Machines.





Maximizing the Margin

The Mathematical Formulation of the "Street Width"

Defining the Margin Width

$$2 / \|w\|$$

The Total Margin (M)

Deriving the Street Width

Assume we have two parallel hyperplanes passing through support vectors:

$$w^T x + b = 1 \quad \text{and} \quad w^T x + b = -1$$

The distance from the origin to plane 1 is $(1-b)/\|w\|$ and to plane 2 is $(-1-b)/\|w\|$. The difference between these two distances yields the total width of the street.

The Constrained Optimization

🎯 **Objective:** We want to maximize the margin $2 / \|w\|$, which is equivalent to minimizing $\frac{1}{2}\|w\|^2$.

🛡️ **Constraint:** For all training points, the point must be correctly classified and stay outside the

street: $y_i (w^T x_i + b) \geq 1$

🔗 **Support Vectors:** Points where $y_i(w^T x_i + b) = 1$ are the only ones that define the boundary.

Mathematical Formulation Comparison

Concept	Primal Problem	Dual Problem
Primary Variable	Weights (w) and Bias (b)	Lagrange Multipliers (λ)
Objective	Minimize $\frac{1}{2} w ^2$	Maximize $\sum \lambda - \frac{1}{2} \sum \sum \lambda y \lambda y (x \cdot x)$
Kernel Handling	Difficult to integrate	Natural fit (uses dot products)
Computation	Scales with Feature Count (D)	Scales with Training Samples (N)

Key Derivation Steps



Formulation

Converting the geometric intent of "widest road" into the quadratic programming problem of minimizing weights.



Lagrangian

Using Lagrange multipliers to combine the objective and constraints into a single differentiable function.



Stationarity

Taking partial derivatives w.r.t w and b to find conditions for the global minimum in the convex space.

“SVM doesn't just separate the data; it finds the street with the maximum buffer to ensure maximum generalization on unseen points.”

— *Mathematical Interpretation of SVM Intuition*

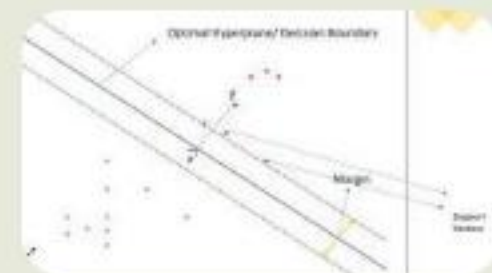
Questions?

Deep dive into SVM Math Intuition complete.

Source: CampusX 100 Days of Machine Learning

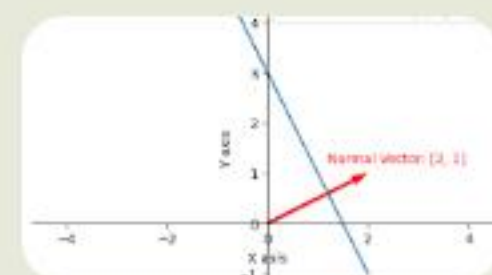
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